

A Short Report on Simulation-Based Inference

Scientific Machine Learning Seminar

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Abstract

Simulation-Based Inference (SBI) offers a novel approach to solve inverse problems powered by machine learning. Hereby, a neural network learns to map observed data to a posterior distribution over model parameters. By using simulations, SBI does not require any likelihood evaluation which otherwise often is computationally costly or has to be approximated. This can allow SBI to outperform classical Bayesian inference methods in inference speed and posterior accuracy. However, common problems like model misspecification, neural inaccuracies or miscalibrated posteriors can limit the usefulness of SBI if they are not circumvented. This report presents basic SBI methodology, in particular Neural Posterior Estimation, discusses its advantages and limitations and reviews two exemplary applications for SBI: detecting planetary events from gravitational microlensing and inferring black hole merger parameters from gravitational waves.

1 Introduction

In computational physics there are usually two types of problems to solve: Firstly, forward problems where the parameters and/or boundary conditions of a model are given and the goal is to find data or a data distribution which adheres to this model. For example in the case of given parameters and initial values this is a typical simulation.

Though secondly, there are also inverse problems where the data is the preset and now the parameters or parameter distributions of the assumed model have to be inferred. So here the data likely comes from an observation and the parameters which represent physical properties of the system have to be computed. This is the type of problem that will be explored in this report.

A rather simple case of solving an inverse problem is the use of linear regression to fit a function to a given dataset. But for more complex models also methods with higher efficiency are needed. The classical approach here is to use Bayesian methods to infer the parameter posterior distribution, but Simulation-Based Inference introduces machine learning techniques to solving inverse problems. SBI is able to solve these problems in some cases more efficiently and computationally cheaper. Applications where SBI shows advantages are (but not limited to) gravitational microlensing ([Zhang et al., 2021](#)), gravitational wave physics ([Dax et al., 2023](#)) and neuroscience ([Deistler et al., 2025](#), Sec. 4.2).

In Section 2 the general concepts of classical Bayesian inference are first explored, specifically focussing on Markov-Chain-Monte-Carlo methods. Then SBI and its underlying concepts are

introduced in Section 3 and exemplified in Section 4 with the help of a toy-example. The advantages and limitations of SBI are presented in Section 5. Two exemplary applications with results are portrayed in Section 6 and a conclusion is given in Section 7.

2 Classical Bayesian Inference

The most outstanding feature of Bayesian Inference in comparison to regression is that the output of BI is a posterior distribution of the parameters instead of just the most likely parameter set sampled from this distribution. So instead of just computing $\theta \sim p(\theta|x)$, the result is the full $p(\theta|x)$ distribution. The naming convention here is that θ is the set of parameters from the model and x is the observed data.

The underlying concept is the Bayes equation

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)}p(\theta) \quad (1)$$

from Bayesian statistics where $p(\theta|x)$ is the posterior distribution, $p(x|\theta)$ the likelihood, $p(x)$ the evidence and $p(\theta)$ the prior distribution. While the prior can be chosen adequately according to assumed distributions of the model parameters and the evidence is just a normalizing factor, the likelihood $p(x|\theta)$ has to somehow be calculated, evaluated or approximated in numerical methods of inferencing the posterior. This is the most important and also most costly step of classical Bayesian Inference (Deistler et al., 2025).

Different methods of Bayesian inference solve this in different ways. Markov chain Monte Carlo (MCMC) methods are relatively simple and widely used for inferring posteriors. These evaluate the likelihood many times randomly in the parameters space and build a posterior distribution from that. In Metropolis-Hastings, a MCMC method, the likelihood value at a random point in the parameter space determines if this point is accepted or rejected. The accumulated distribution of the accepted points then converges to the posterior after many evaluations (Prado et al., 2026).

There are adjusted methods that can make the algorithm more efficient (like introducing random walk instead of true random sampling) but all classical Bayesian inference methods suffer from similar key limitations: The convergence to a posterior that is accurate enough for the specific application takes many evaluations of the likelihood and is therefore very computationally expensive. For the applications that are later shown in Section 6 this can be up to 1 second per evaluation, for an adequately sampled posterior distribution the computation time per analyzed dataset can be huge.

In complex models the exact evaluation of the likelihood is often not possible or a lot more computationally expensive, so it has to be approximated by a Gaussian distribution, usually as part of the χ^2 method. This inserts another error source into the posterior distribution.

Classical Bayesian inference methods also have the problem that they cannot model multimodal posteriors very well. The algorithm often gets stuck in one mode and is unable to jump to another, this becomes a problem in models with a lot of close degeneracies (Deistler et al., 2025). In turn, grid searches and human intervention are often needed which increases the time and effort of the data analysis process. This may not be a possibility for the huge amounts of data that needs to be processed in modern science.

3 Introduction to Simulation-Based Inference

The key feature of Simulation-Based Inference is the skipping of the computationally expensive likelihood evaluation. This is the reason SBI is sometimes also called Likelihood-Free Inference. Instead of explicitly using a method to evaluate the likelihood at specific points in parameter space, the likelihood is implicitly given when sampling data from a simulator of the model. A neural network will train on this simulated dataset to infer a parameter posterior distribution from an observation. The likelihood still exists inside the dataset but does not have to be calculated or approximated.

This report focusses on the direct inference of posteriors, this methodology is called Neural Posterior Estimation (NPE). This is in contrast to Neural Likelihood Estimation (NLE) and Neural (Likelihood-)Ratio Estimation (NRE), they respectively require an additional step of computation to achieve the posterior but may offer advantages or more insight in the analysis process.

The SBI workflow can be subdivided into three steps as shown in Fig. 1: First, the simulator has to generate a relatively large dataset in accordance with the physical model. To achieve this, an adequate parameter prior distribution is chosen according to guesses on the physical properties of the model, e.g. the range of specific parameters. The simulator then samples x_i from the likelihood $p(x|\theta_i)$ and therefore generates a (x_i, θ_i) dataset. The likelihood is implicitly contained in the dataset but because the simulation solves a forward problem, this step is not very computationally costly.

Next, a neural network is trained to map each tuple (x_i, θ_i) to an estimated posterior density $q_\phi(\theta_i|x_i)$. The network architecture can in theory be chosen freely but Normalizing Flows is commonly used. The loss function

$$L(\phi) = E_{()} \quad (2)$$

is minimized to improve the network weights. It consists of the expectation value over all sampled tuples of the negative logarithm of the estimated posterior. This is very similar to the definition of a cross entropy and will result in a balance between maximizing $q_\phi(\theta_i|x_i)$ at the correct θ_i and confidence or slimness of the distribution. So the estimated posterior density $q_\phi(\theta_i|x_i)$ converges to the true posterior $p(\theta_i|x_i)$.

Lastly, the observed datapoint x_{obs} is inserted into the network and evaluated inside an interval of θ at any resolution. This will output the posterior distribution of any observation with relatively low computational cost. There is no need for further training when analyzing new data under the assumption of the same model.

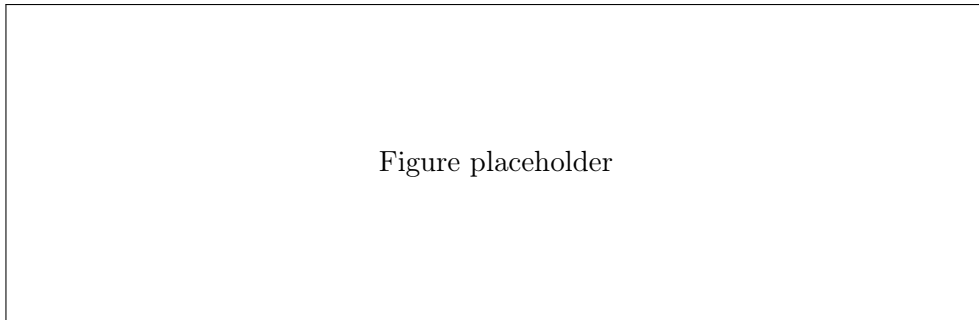


Figure 1: Workflow of SBI.

4 Simple SBI Example

A fairly simple demonstration of SBI is done here with the help of the example of an angled throw.

5 Benefits and Limitations

Key benefits: very fast inference (Amortized Inference), Likelihood-free (no approximation or expensive computation), multimodal posteriors

Limitations: Upfront training, model misspecification (common pitfall) posterior is a neural estimation uncalibrated posterior

6 Application

Short explanation of microlensing

Results of usage of SBI (Zhang et al., 2021)

Short explanation of gravitational waves and black hole mergers

use of flow matching networks

results of usage here (Dax et al., 2023)

Other applications

7 Conclusion

Benefits and Limitations (stuff from abstract)

When to use SBI inverse problem, time critical inference, expensive likelihood evaluation, multimodal posterior

References

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